H.M. SARWER ALAM

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Objective

Computer Science graduate and Software Engineer with a strong foundation in Python, machine learning, and research-driven problem solving. Interested in solving new and meaningful problems through practical AI/ML development, experimentation, and innovation, with a focus on building efficient and reliable solutions for real-world challenges.

Experience

- **BNX Network Inc.** [] — *April 2025 – Present*
  Software Engineer (Junior) (formerly Trainee Software Engineer) — Naperville, IL, USA; Dhaka, Bangladesh
  Responsibilities
  - Developed and deployed ML/AI solutions, including model training (Random Forest, XGBoost, SVM) and backend services using FastAPI and custom TCP-based server-client architecture.
  - Worked on research-driven and task-based problem solving in traditional AI/ML systems, including experimentation, implementation, and performance improvement.
  - Built web applications using **Python (Django)**, HTML, and CSS; collaborated on **C# .NET** development and participated in testing, validation, and debugging processes.
- **BRAC University** [] — *May 2023 – April 2024*
  Student Tutor (Undergraduate Teaching Assistant), Computer Science and Engineering — Dhaka, Bangladesh
  Responsibilities
  - Provided 12-hour weekly tutoring on programming, debugging, and problem-solving; delivered personalized guidance and study support.
  - Evaluated lab assignments with constructive feedback to promote continuous student improvement.

Education

- **BRAC University** — *Session: 2020 - 2024*
  Bachelor of Science, Computer Science and Engineering — Dhaka, Bangladesh
  - CGPA: 3.88/4.00
  - Vice Chancellor's List for 4 semesters
  - Dean's List for 5 semesters
- **Bir Shreshtha Noor Mohammad Public College** — *Passing year- 2019*
  Higher Secondary School Certificate (HSC) — Dhaka, Bangladesh
  - GPA: 5.00/5.00
- **Bir Shreshtha Noor Mohammad Public College** — *Passing year- 2017*
  Secondary School Certificate (SSC) — Dhaka, Bangladesh
  - GPA: 5.00/5.00

Projects

- **CiteWeave-DuckSummarizer** — *Personal Project*
  Tools: Python, BeautifulSoup, SentenceTransformer, TF-IDF, DuckDuckGo Search []
  Developed an intelligent web summarization tool that fetches and ranks web content using hybrid summarization (TF-IDF + embeddings) and cosine similarity filtering to generate concise, cited summaries in console and HTML formats.
- **Predicting Heart Disease Using Machine Learning** — *University Individual Project*
  Tools: Python, scikit-learn, Pandas, Matplotlib, Logistic Regression, Decision Trees, KNN, SVM, Kaggle dataset []
  Implemented Logistic Regression, Decision Trees, KNN, SVM, and Naive Bayes for prediction and applied data preprocessing using Pandas and NumPy for features and dataset preparation.
- **Ask My PDF – Local RAG Chatbot** — *Personal Project*
  Tools: Python, FAISS, Sentence Transformers, PyTorch, Transformers, TinyLlama []
  Built a fully offline chatbot enabling natural language interaction with PDF documents using FAISS-based retrieval, local LLM inference (TinyLlama), and a complete RAG pipeline for text extraction, embedding, and chunking, with planned Streamlit UI and multi-PDF support.
- **Random Image Classification** — *University Group Project*
  Tools: Python, ResNet50, AlexNet, Inception Net, Custom Model []

RESEARCH AND PUBLICATION

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[J.1 & T.1] Farhan Faruk, **H. M. Sarwer Alam**, Rafeed Rahman, Md. Golam Rabiul Alam, Junho Jeong, Md. Kabir Hossain, and Jia Uddin. (2025). **RenSeg: Leveraging Unsupervised Segmentation using Localization and Contour-Guided Quickshift for Renal Calculi and Carcinoma Segmentation and Classification**. *IEEE Journal of Biomedical and Health Informatics (JBHI)*, IEEE Xplore. **DOI:**10.1109/JBHI.2025.3629580.
Available at: https://ieeexplore.ieee.org/document/11286221. **Impact Factor:** 6.8
Contribution: Designed an unsupervised medical image segmentation pipeline ensuring accuracy and reliability in clinical applications.

[J.2] Tasnim Sakib Apon, Md. Golam Rabiul Alam, Md. Tanzim Reza, Sangita Baidya, Mohammad Fahim Tahmid, Md. Ashraful Alam, Farhan Faruk, **H. M. Sarwer Alam**, Muhammad Almoyad, Khondokar Fida Hasan, and Mohammad Ali Moni. (2026). **LeukocyteNet: An Explainable Transfer-Transformer Fusion Learning Model for Leukocyte Classification**. *Emerging Science Journal*, 10(3), 1239–1262.
DOI: 10.28991/ESJ-2026-010-03-03. **Available at:** https://ijournalse.org/index.php/ESJ/article/view/3459.
Impact Factor: 5.428
Contribution: Co-developed a hybrid CNN-Transformer model with explainable AI for leukocyte classification.

[C.1 & S] Rakib Hossain Rifat, Farhan Faruk, H. M. Sarwer Alam, Jingjing Yao, and Md. Golam Rabiul Alam. (2025). **PLFL: A Semi-supervised Federated Learning Approach Leveraging Pseudo-labeling**. *European Conference on Computer Vision (ECCV) 2026, European Computer Vision Association*.
Status: Submitted. **Contribution:** Proposed PLFL, a privacy-preserving federated semi-supervised model achieving 98.34% accuracy on BloodMNIST.

SKILLS

- **Programming & Tools:** Python, SQL, C, C#, HTML, CSS, FastAPI, Django, TensorFlow, PyTorch, scikit-learn scikit-learn, Microsoft Office, Ubuntu, CUDA, Docker (Basic), Github
- **Core Competencies:** ML Model Development & Deployment,, Problem Solving, Teamwork, Leadership, Communication, Adaptability

KEY LEARNING AREAS

Interested in AI, ML, and Image Processing; open to learning and contributing to emerging areas like NLP and Quantum Machine Learning.

SCHOLARSHIP AND CERTIFICATION

- **Best Thesis Award - Spring 2024** *Feb 28, 2025*
  Brac University
  Received the "Best Thesis Award" for the work titled *"Leveraging Unsupervised Segmentation for Semi-supervised Renal Calculi and Carcinoma Classification"* from the Department of CSE, Brac University.

EXTRA CURRICULAR ACTIVITIES

- **Senior Executive - Communications and Marketing** *August, 2021 - August, 2023*
  BRAC University Computer Club
  Led and organized communication and marketing efforts for club events and activities.
- **House Vice Captain - MJ House** *January, 2018 - April, 2019*
  Bir Shreshtha Noor Mohammad Public College
  Oversaw house activities and helped organize events and competitions.

REFERENCES

1. **Md Golam Rabiul Alam, PhD**
   Professor of Computer Science & Engineering
   BRAC University
   Email: rabiul.alam@bracu.ac.bd
   Phone: +8801797347635
2. **Zahin Ahmed**
   Graduate Research Assistant (PhD student in CE)
   The George Washington University, Washington D.C
   Lecturer (on-leave) at Brac University, Dhaka.
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